

UID EX 10

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AIM:

The aim is to create data visualizations, such as pie charts and bar graphs, for an inventory management system using JavaScript.

PROCEDURE:

Step 1:

Set Up Your HTML File

```
<> INDEX.HTML > html
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4  <meta charset="UTF-8">
5  <meta name="viewport" content="width=device-width, initial-scale=1.0">
6  <title>Student Performance Visualization</title>
7
8  <style>
9  body {
10     font-family: Arial;
11     text-align: center;
12     margin: 50px;
13 }
14 canvas {
15     margin: 20px auto;
16 }
17 </style>
18
19 </head>
20 <body>
21
22 <h1>Student Performance System</h1>
23
24 <canvas id="pieChart" width="400" height="400"></canvas>
25 <canvas id="barChart" width="400" height="400"></canvas>
26
27 <script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
28 <script src="script.js"></script>
29
30 </body>
31 </html>
```

Step 2:

Create the JavaScript File for Charts Next, create a JavaScript file (script.js) to handle the data visualization logic.

```

JS script.js > ...
1  // Data for student performance
2  const studentData = {
3      labels: ['Math', 'Science', 'English', 'Computer', 'History'],
4      datasets: [{
5          label: 'Marks Obtained',
6          data: [85, 90, 75, 95, 80],
7          backgroundColor: [
8              '#FF6384',
9              '#36A2EB',
10             '#FFCE56',
11             '#4BC0C0',
12             '#9966FF'
13         ]
14     }]
15 };
16
17 // PIE CHART (Subject-wise distribution)
18 const ctxPie = document.getElementById('pieChart').getContext('2d');
19
20 new Chart(ctxPie, {
21     type: 'pie',
22     data: studentData,
23     options: {
24         responsive: true,
25         plugins: {
26             title: {
27                 display: true,
28                 text: 'Marks Distribution by Subject'
29             }
30         }
31     }
32 });
33
34 // BAR CHART (Marks comparison)
35 const ctxBar = document.getElementById('barChart').getContext('2d');
36
37 new Chart(ctxBar, {
38     type: 'bar',
39     data: studentData,
40     options: {
41         responsive: true,
42         plugins: {
43             title: {
44                 display: true,
45                 text: 'Marks Comparison'
46             }
47         },
48         scales: {
49             y: {
50                 beginAtZero: true
51             }
52         }
53     }
54 });

```

OUTPUT:

Student Performance System

